```
import zipfile
import os

zip_path = "/content/archive (2).zip"
extract_path = "/mnt/data/insurance_ridge_data"

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall(extract_path)

os.listdir(extract_path)
```

```
['insurance.csv']
```

```
for root, dirs, files in os.walk(extract_path):
    for file in files:
        if file.endswith(".csv"):
            print("CSV Found:", os.path.join(root, file))
```

```
CSV Found: /mnt/data/insurance_ridge_data/insurance.csv
```

```
import pandas as pd

csv_path = "/mnt/data/insurance_ridge_data/insurance.csv"   # change if needed
df = pd.read_csv(csv_path)

df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

```
df_encoded = pd.get_dummies(df, columns=["sex", "smoker", "region"], drop_first=True)
df_encoded.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

```
X = df_encoded[["age", "bmi", "children", "smoker_yes"]]
y = df_encoded["charges"]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
from sklearn.linear_model import LinearRegression

lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)
```

```
from sklearn.linear_model import Ridge

alphas = [0.1, 1, 10, 100]
ridge_preds = {}

for a in alphas:
    ridge = Ridge(alpha=a)
    ridge.fit(X_train, y_train)
    ridge_preds[a] = ridge.predict(X_test)
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

def evaluate(y_test, y_pred):
    mae = mean_absolute_error(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)
    return mae, rmse, r2
```

```
results = {}

# Linear Regression
results["Linear Regression"] = evaluate(y_test, y_pred_lr)

# Ridge Regression
for a in alphas:
    results[f"Ridge alpha={a}"] = evaluate(y_test, ridge_preds[a])

results_df = pd.DataFrame(results, index=["MAE", "RMSE", "R2 Score"]).T
results_df
```

	MAE	RMSE	R2 Score
Linear Regression	4213.798595	5829.378522	0.781115
Ridge alpha=0.1	4214.995041	5829.680380	0.781092
Ridge alpha=1	4225.750279	5832.609778	0.780872
Ridge alpha=10	4330.796360	5880.507334	0.777258
Ridge alpha=100	5204.424227	6951.187294	0.688764

```
best_model = results_df["RMSE"].idxmin()
print("✅ Best Model based on Lowest RMSE:", best_model)
```

```
✅ Best Model based on Lowest RMSE: Linear Regression
```